



Part II

Foundations of AI

and ML in

Algorithmic

Trading

Artificial intelligence (AI) refers to using advanced computational techniques and algorithms to perform complex tasks that typically require human intelligence. These tasks range from recognizing patterns in data and understanding natural language to making data-driven decisions.

While machine learning (ML) and AI are closely related fields, they are not synonymous: ML is a subset of AI that deals with developing algorithms and statistical models enabling computers to learn and make decisions without being explicitly programmed for specific tasks. AI includes other algorithms, such as rule-based systems and symbolic reasoning, which do not necessarily involve learning from data.

One of the critical applications of AI in finance is algorithmic trading, with AI analyzing vast amounts of historical data for profitable patterns, making predictions about future market movements with complex market dynamics models, and executing trades at the speed, scale, and efficiency superior to any human trader.

In addition, financial markets are inherently complex and unpredictable, and AI models must be carefully designed and trained, which includes selecting the right data sets, choosing the appropriate algorithms, and continuously monitoring and refining the models to improve their accuracy and reliability.

This part has been designed to provide a practical introduction to navigating this environment. We start with the outline of the step-by-step guide for AI-based financial forecasting, and then discuss each step in detail with hands-on examples illustrating the industry's best practices.

Step-by-step Guide for AI-based Algorithmic Trading

Step 1: Problem Definition

1. Define the problem and the objective; choose the variable you intend to predict or optimize.

Step 2: Dataset Preparation

1. Collect relevant data.
2. Carry out exploratory data analysis.
3. Preprocess the data.
 1. Clean the dataset of missing values.
 2. Clean the dataset of outliers.
 3. Carry out feature engineering.
 4. Consider normalization or standardization of certain features.

5. Consider transforming time series features to be stationary.
6. Consider identifying cointegrated time series with the Engle-Granger test.
4. Select the features, i.e., the independent variables with the strongest relationship to the dependent variable (the objective). Use the following:
 1. Correlation analysis
 2. Features importance analysis
 3. Auto-identification of features
 4. Dimensionality reduction/principal component analysis whenever possible
5. Split the dataset into a training and test dataset.

Step 3: Model Choice, Training, and Application

1. Select the most appropriate forecasting algorithm for the model.
2. Train the model typically, by adjusting the model parameters to best fit the historical training dataset, by minimizing the error between the predicted and actual values.
3. Evaluate the model on the test dataset using metrics, such as R-squared or mean square error (MSE). If needed, use cross-validation.
4. Deploy the model and keep retraining it for updated data sets.